

Exploring Touch Interactions for Input Visualization in Personal Informatics

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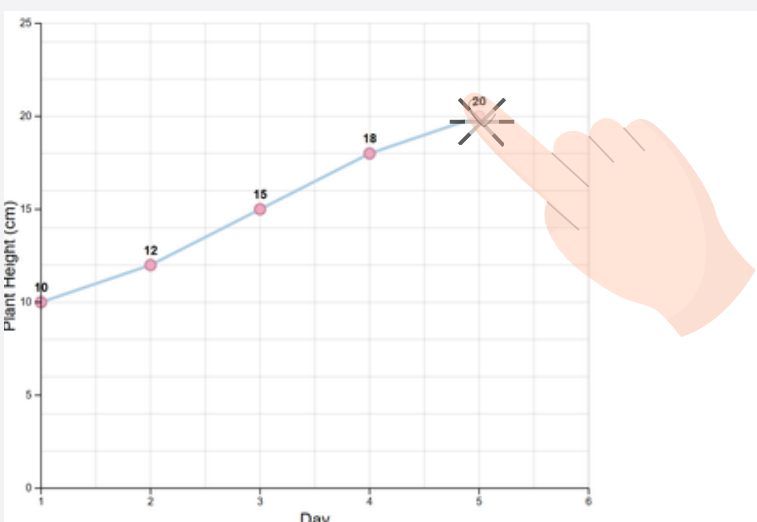
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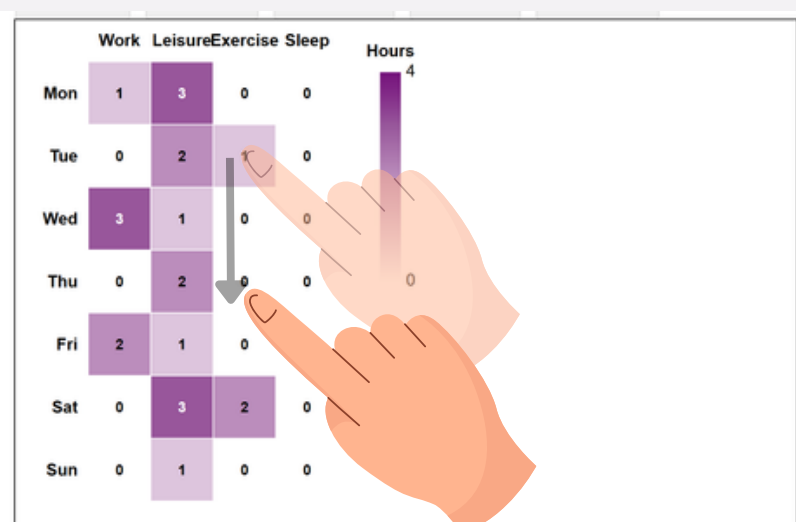
INTRODUCTION

Input visualizations enable people to not only view but also to manipulate data directly through a visualization. We explore what types of interaction techniques people expect or prefer when inputting data through visualizations on touchscreens specifically in the context of personal informatics.

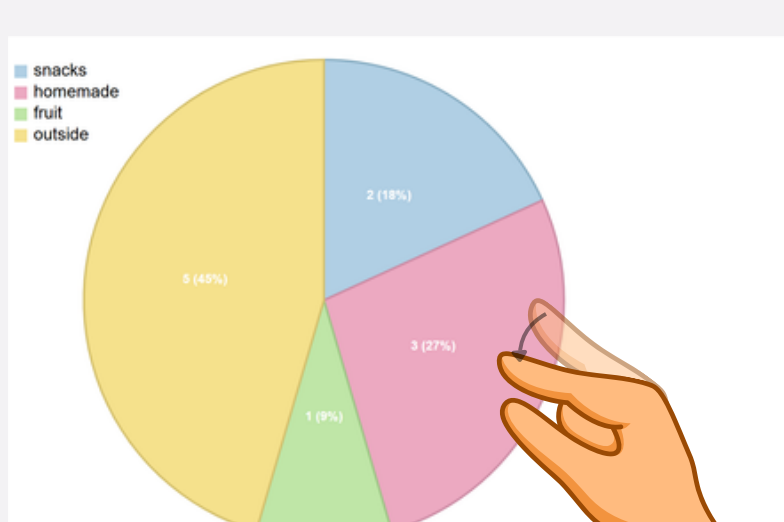
Examples of explored gestures **5 operations** across **5 Idioms**



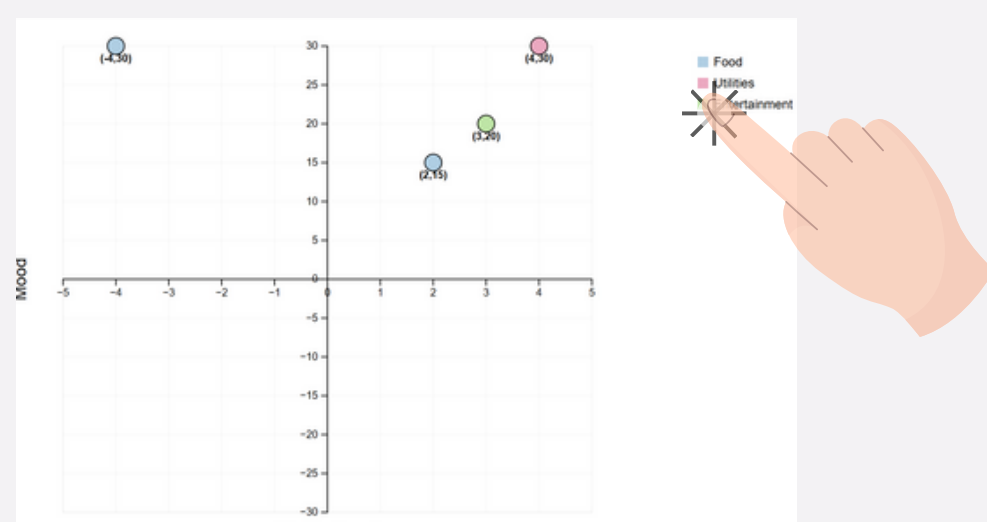
Tap to **add** new point on the **line**



swipe down to **remove** from **Heatmap** cell



Pinch to **change** **pie** section



long press legend to **add** **category** to **scatter** plot



Drag out to **remove** **category** from the **bar** chart

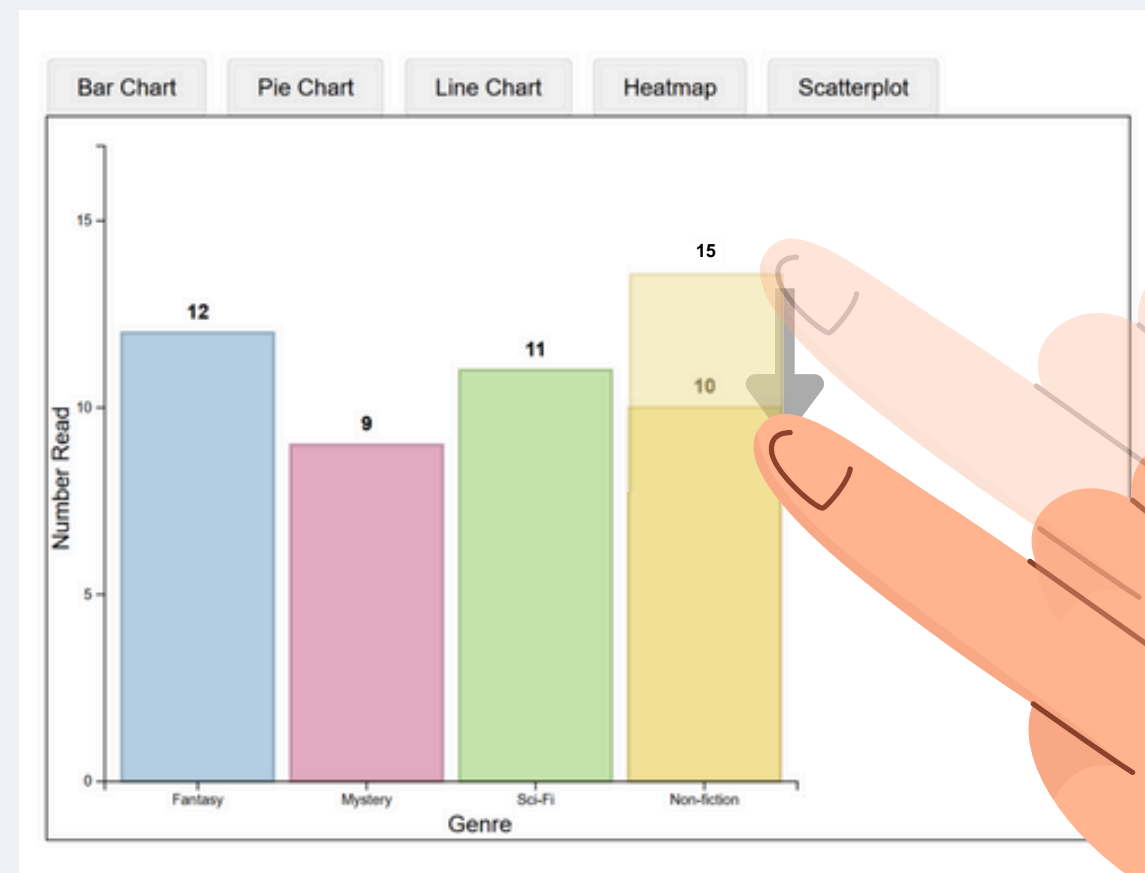
STUDY



For each idiom participant suggests a gesture using the canvas, data marks, GUI, or other elements

Gesture Assignments			
	barArea	barTopEdge	outsideBars
barAddToBar	tap	select	select
barRemoveFromBar	select	select	select
barChangeBar	pan	select	select
barAddBar	select	select	select
barRemoveBar	select	select	select
barMergeBars	double tap	select	select
barReorderBar	hold	select	select
	pinch	select	select
	pinch in	select	select
	pinch out	select	select
	swipe left		
	swipe right		

Facilitator modifies the prototype to implement the interaction suggested by the participant



Participant interacts with the prototype again to try out their suggested gesture

We had 6 participants. Their screen, hand, and voice was recorded and used for data analysis.

FINDINGS

Direct Manipulation

- Lower operation complexity (like add, remove and change) prompts direct manipulation gestures
- Higher GUI usage of pie chart suggest less ways to manipulate pie chart data directly
- Add/remove category elicited more interactions with the legends/axis
- Data encoded by color (like heatmap) had no suggested direct manipulation interaction to change the color

Operation Type, Symetry and reversibility

- Operations requiring granular control (change) prompted continuous gestures
- Operation symmetry (Add - Remove) prompted similar gestures for that idiom
- Expectation of shorter gestures for the add and remove operation

Visual Encodings

- Idioms encoding data by area (bar, pie) share gestures; so do those encoding data in space (scatter, line)

Idioms	Operations	Mark					Interaction					GUI					Data		Steps		
		canvas	data mark	Extensive data mark	GUI	legend and axis	press	tap	drag	pinch	complex	slider	tool	menu	button	text field / Form	discreet	continuous	1	2	3
Bar	Add	0	2	0	0	0	0	2	0	0	0	0	0	0	0	0	2	0	2	0	0
Pie	Add	0	4	0	0	1	2	2	0	1	0	0	0	1	0	0	4	1	4	0	1
Line	Add	6	1	0	1	0	0	6	2	0	0	0	0	0	0	0	8	0	7	1	0
Heat	Add	0	6	0	0	0	0	6	0	0	0	0	0	1	0	0	6	0	5	0	1
Scatter	Add	3	2	0	0	7	0	11	1	0	0	0	0	2	0	0	11	1	1	6	2
Bar	Remove	0	2	0	0	0	0	2	0	0	0	0	0	0	0	0	2	0	2	0	0
Pie	Remove	0	0	0	2	0	0	2	0	0	0	0	0	0	0	0	2	0	2	0	0
Line	Remove	0	13	0	0	0	6	2	3	1	1	0	0	5	0	0	8	5	6	4	3
Heat	Remove	0	4	0	0	0	2	2	0	1	0	0	0	0	0	0	4	0	4	0	0
Scatter	Remove	0	10	0	0	0	5	3	2	0	0	0	0	3	0	0	8	2	5	2	3
Bar	Change	0	8	3	0	1	0	5	7	0	0	0	0	1	0	1	5	7	9	0	3
Pie	Change	0	15	0	3	3	1	12	5	3	0	0	0	4	4	0	14	7	12	3	6
Heat	Change	1	8	0	1	3	0	5	8	0	0	1	0	1	1	0	8	5	8	4	1
Line	Change height	0	8	2	0	0	1	4	5	0	0	2	0	0	0	0	5	5	7	1	2
Scatter	Change pos	0	6	0	0	0	0	1	5	0	0	0	1	0	0	0	1	5	5	1	0
Scatter	Change Cat	0	0	0	1	7	2	2	4	0	0	0	1	0	0	0	5	3	3	5	0
Bar	Add Category	3	2	0	4	6	2	5	8	0	0	0	0	1	4	0	7	8	13	2	0
Pie	Add Category	0	2	1	2	5	0	5	4	0	1	0	0	1	1	0	9	1	8	1	1
Scatter	Add Category	0	0	0	3	2	0	5	0	0	0	0	0	0	3	0	5	0	5	0	0
Bar	Remove Category	2	6	0	0	4	3	0	8	1	0	0	0	2	0	0	3	9	7	5	0
Pie	Remove Category	0	8	0	0	5	9	2	2	0	0	0	0	6	0	0	13	0	2	5	6
Scatter	Remove Category	0	0	0	0	5	2	1	2	0	0	0	0	1	0	0	5	0	4	1	0

Table: Frequency of suggested interactions per operation across identified codes